OPERATING SYSTEMS

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File Management

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Access Matrix

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Slides Credits for all PPTs of this course



- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau

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Access Matrix



Protection Model

- The Access Matrix is a protection model used by an operating system to specify who can access what and what operations are permitted.
- It represents access permissions between:
  - Domains → Subjects/users/processes
  - Objects → Resources such as files, printers, memory, etc.
- Each cell contains the access rights allowed for a particular domain on an object.

Basic Structure

- **Rows → Domains**
- **Columns → Objects**
- **Cells → Access Rights**

Example Rights - read (r), write (w), execute (x), print, delete

Access Matrix example



- Domains: **D1, D2, D3, D4**
- Objects: **F1, F2, F3, Printer**

Key Points:

- **D1** → Read F1, F3
- **D4** → Same as D1 + Write F1, F3
- **D2** → Access Printer only

object / domain	F_1	F_2	F_3	printer
D_1	read		read	
D_2				print
D_3		read	execute	
D_4	read write		read write	

Access Matrix



Ensures:

- Only **allowed access**
- Strict control of resources

OS decides:

- Which domain a process runs in
- What rights are given

Domain Switching

- Domain is treated as an **object**
- Switching = performing **switch operation**

Rule:

- Process can switch from **Di → Dj**
  only if 'switch' right exists

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Access Matrix – Domain switching



- D2 → can switch to D3, D4
- D4 → can switch to D1
- D1 → can switch to D2

Additional Rights:

- **copy** → duplicate access rights
- **owner** → full control
- **control** → modify rights

object / domain	F_1	F_2	F_3	laser printer	D_1	D_2	D_3	D_4
D_1	read		read			switch		
D_2				print			switch	switch
D_3		read	execute					
D_4	read write		read write		switch			

Implementation of the Access Matrix



Why is Implementation Needed?

- The Access Matrix provides a clear protection model, but a complete matrix can become very large.
- Most entries in the matrix are usually empty.
- Therefore, storing the entire matrix directly is not memory efficient.
- Operating systems use different techniques to implement the Access Matrix.

Common Implementations

1. **Global Table**
2. **Access List for Objects (ACL)**
3. **Capability List**

Implementation of the Access Matrix – Global table



- A global table stores all non-empty access relationships in a single table.
- Each entry is represented as: **<Domain, Object, Rights>**

How It Works

1. A process requests access to an object.
2. The OS searches the global table.
3. It looks for the corresponding Domain–Object entry.
4. The requested operation is checked against the stored rights.
5. Access is allowed or denied.

Limitations

- **Large table** → inefficient use of memory
- **Slow lookup** → searching may take time
- **Difficult to manage grouping** of similar access rights

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Implementation of the Access Matrix – Access list for objects

- In this approach, each object maintains a list of domains and their allowed rights.
- The list answers: **“Who can access this object, and what can they do?”**
- Format: Object → (Domain, Rights)

How it works

1. Process requests access to an object.
2. OS checks that object's access list.
3. It finds the requesting domain.
4. The requested operation is compared with the allowed rights.
5. Access is granted or denied.

Advantages

- Easy to determine who can access an object.
- Permissions can be easily modified.
- Supports default permissions.

Limitation - Difficult to determine all objects accessible by a particular domain.

Domain	Rights
D1	Read
D3	Read, Write
D4	Read, Write

So:

- D1 → can **read** F1
- D3 → can **read and write** F1
- D4 → can **read and write** F1
- D2 → has **no access** to F1

Capability List



- A capability list stores the access rights associated with a domain.
- It answers: **“What objects can this domain access, and what can it do?”**
- Format: Domain → (Object, Rights)

How it works

1. A process executes in a particular domain.
2. The OS checks that domain's capability list.
3. It finds the requested object.
4. The requested operation is checked against the stored rights.
5. Access is granted or denied.

Object	Rights
F1	Read
F2	Read, Write
Printer	Print

So D1 can:

- **Read** F1
- **Read and Write** F2
- **Print** using the Printer



THANK YOU

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